

Digitisation Errors and Hamiltonian Truncation Effective Theory in 0+1D Scalar Field Theory for Quantum Computation

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The origin, structure and mitigation of digitisation errors in quantum simulations are investigated for 0+1 dimensional scalar field theories using harmonic and anharmonic oscillator systems. Following the work of Klco and Savage, we analyse three digitisation schemes: Jordan-Lee-Preskill (JLP), its finite-difference corrected (FDC) variant and a harmonic oscillator (HO) basis. We reproduce their established results for digitisation error scaling and demonstrate the role of the Nyquist-Shannon sampling constraints in controlling these errors. We extend this analysis using Hamiltonian Truncation Effective Theory (HTET) within Cohen *et al.* as a method for compensating for truncation effects arising from finite dimensional Hilbert space representations. By studying ground state energies within an interacting scalar field theory, we show HTET can reduce the required Hilbert space dimension, and therefore qubit requirements, for a given target accuracy within specific parameter regions. However, we identify regions exhibiting pronounced instability bands in which higher order corrections exhibit regions of numerical instability. Through diagnostics we find these features are not driven by small energy denominators, large matrix elements or a breakdown of perturbative control. Instead we suggest these features arise from the dependence of the truncated Hilbert space on the HO basis tuning parameter, where discrete changes in the retained states lead to enhanced sensitivity near the truncation boundary.

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1. INTRODUCTION

Quantum simulation of quantum systems is regarded as one of the most promising applications of quantum computing. Unlike classical computers, which are forced to deal with exponential scaling when representing quantum states, quantum processors generalise naturally for representing quantum states, as qubit Hilbert spaces scale exponentially with system size. This has motivated the application of quantum algorithms for problems within chemistry, material science, optimisation and other physics problems [1–4]. The use of quantum computing has become more prevalent than it ever has before [5] and although quantum advantage over classical computation has not yet been universally verified [6–9], improvements in functionality and qubit counts develop at a rapid pace [10]. The noisy-intermediate-scale-quantum (NISQ) era, coined by John Preskill [11], refers to the current period of quantum computing where most circuit based quantum processors contain a number of qubits on the scale of $10^1 - 10^2$ and have a distinct absence of fully fault-tolerant error correction [12, 13].¹ Given this, practical quantum simulations within the NISQ era must be designed to minimise resource requirements while controlling sources of systematic and hardware induced error [15].

A central challenge arises when performing quantum simulations of field theories, namely representation of continuous degrees of freedom using a finite number of qubits. Scalar field theories form a fundamental class of models in quantum field theory (QFT) and are defined over an infinite dimensional Hilbert space. To implement such systems on a quantum computer, the field must be both discretised and truncated thus reducing a continuum theory to a finite dimensional representation. This process is commonly referred to as digitisation. It introduces systematic errors that are independent of hardware noise but instead arise from the approximation to a finite discrete basis. Understanding the errors introduced by digitisation is therefore a prerequisite for reliable quantum simulation of quantum field theories on NISQ devices.

To study these effects, a useful setting is provided by 0+1 dimensional scalar field theory, which is equivalent to quantum mechanics of a single degree of freedom. In this limit we are able to study non interacting and interacting scalar field theories through harmonic and anharmonic oscillators, whose spectra and wave functions are well understood. Direct comparison between numerically obtained results and exact continuum solutions are able to isolate digitisation errors without additional complications from spatial discretisation or many body effects [16].

Multiple approaches to digitising scalar fields have been developed such as the widely used Jordan-Lee-Preskill (JLP) basis. Within the JLP basis the field is directly discretised in position space over a finite interval and represented on a uniform grid and the conjugate momentum is implemented via a quantum Fourier transform which enforces the canonical relationship between field and conjugate momentum representations [17]. An alternative approach within this basis replaces the Fourier relation with a finite difference approximation, this introduces polynomial discretisation errors but simplifies implementation. In grid based methods such as JLP,

¹ These are not to be confused with quantum annealers which contain a number of qubits on the scale of 10^3 at present [11, 14] and are not suited to diverse problem sets and whose computational advantages still lack a theoretically compelling arguments so we do not investigate them here.

errors due to digitisation arise from the finite domain and resolution of the sampling grid along with the constraints introduced by the Fourier relation between field and momentum variables. These competing effects can be understood within the context of the Nyquist-Shannon sampling theorem which provides conditions under which errors like these become exponentially suppressed [16, 18].

An alternative strategy to the JLP basis is to use a harmonic oscillator (HO) basis, in which the Hilbert space is truncated in terms of energy eigenstates of a harmonic oscillator Hamiltonian rather than field values. This basis is adapted to systems with localised wave functions and can provide efficient representations of low energy states [16]. The errors introduced within the HO basis originate from the truncation in the energy space and frequency of the HO basis. This frequency determines how well the chosen basis captures the support of the physical wave function which is attempting to be simulated.

The study of QFTs is a central challenge within theoretical physics, particularly in regimes where perturbative methods breakdown. Traditionally, lattice discretisation has provided a framework for investigations however it can be computationally demanding, especially within the realm of real time simulations [19]. An alternative approach to this is Hamiltonian truncation (HT). Within HT the infinite dimensional Hilbert space of a QFT is projected to a finite subspace with a defined energy cut-off. This allows QFTs to be studied without introducing a spatial lattice and is applied as a numerical technique to investigate interacting field theories and their spectra, previous work demonstrates it as an efficient alternative to lattice models [20]. HT has also been considered within the context of quantum computing, where truncated Hamiltonians can be mapped directly onto qubit systems, this offers a potential route to the simulation of QFT dynamics while reducing quantum resource requirements [21]. Implementations such as these typically rely on direct truncation of the Hilbert space, introducing errors which are not systematically controlled and must be reduced through increased size of the truncated basis. In contrast, Hamiltonian Truncation Effective Theory (HTET) is a perturbative matching procedure used to construct an effective Hamiltonian, which is then treated non-perturbatively within the truncated Hilbert space. It treats the truncation as a physical ultraviolet cut-off and incorporates the effects of the excluded high energy states through controlled corrections organised in inverse powers of the truncation boundary in powers of inverse energy truncation boundary (E_{max}). This results in an effective Hamiltonian defined within the truncated space that can systematically improve the naive truncation.

It is then a natural step to the application of this method for the digitisation of QFTs onto qubit systems. HTET has the potential to reduce truncation errors without increasing the size of the Hilbert space. The primary aim of this work then is to determine how digitisation errors in scalar field theory simulations scale with Hilbert space truncation and discretisation, reproducing results from Klco and Savage [16], and then to assess whether HTET provides a method for reducing qubit costs for equivalent errors. To this end we first reproduce and analyse the JLP basis, finite difference corrections and the HO basis within the context of 0+1 dimensional scalar field theory. This allows for systematic benchmarking against known results and provides a consistent framework for comparison. From this we implement HTET within the harmonic

oscillator basis and examine its effectiveness in reducing truncation errors.

Within section 2, we introduce the 0+1 dimensional scalar field theory and the formulation of the studied harmonic and anharmonic oscillators. Section 3 presents the digitisation procedures, including JLP, finite-difference and HO basis, together with an analysis of the associated errors and the role of the Nyquist-Shannon sampling theorem. In section 4 we introduce HTET and derive the corresponding effective Hamiltonians used within this work. Numerical results and comparisons between methods are then presented and discussed within the relevant sections. Finally conclusions and direction for future work are discussed in sections 5 and 6.

2. 0+1 DIMENSIONAL SCALAR FIELD THEORY

As stated within this work we aim to isolate and mitigate the errors caused by the process of digitisation of scalar field theories onto quantum computers. To do this we explicitly calculate observables from a matrix formulation of the Hamiltonian. We examine both an interacting and non-interacting 0+1 dimensional scalar field theory through a harmonic and anharmonic oscillator. These systems have energy spectra which are easily found², allowing digitisation errors to be isolated and quantified.

Following [16], starting from continuum scalar field theory, a generic system describing the dynamics of a scalar field with self interaction is discretised onto a spatial lattice yielding a Hamiltonian expressed as a sum over lattice sites. Restricting this to the 0+1 dimensional case gives the resulting Hamiltonian corresponding to a quantum mechanical oscillator with a quartic self interaction

$$H = \frac{1}{2}\Pi^2 + \frac{1}{2}m_0^2\phi^2 + \frac{\lambda_0}{4!}\phi^4 \quad (1)$$

where ϕ and Π are the field and conjugate momentum representations respectively, m_0 is the oscillator mass and λ_0 is the self interaction coefficient. To simplify numerical treatment the field and conjugate momentum are rescaled to dimensionless variables $\bar{\Pi}, \bar{\phi}$ by appropriate powers of the mass, so that the explicit mass dependence from the quadratic term is removed. This preserves the canonical commutation relation, $[\bar{\phi}, \bar{\Pi}] = i$. This rescaling yields the standard harmonic oscillator Hamiltonian and the interacting Hamiltonian:

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2 \quad (2)$$

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2 + \frac{\lambda_0}{4!}\bar{\phi}^4 \quad (3)$$

These dimensionless Hamiltonians form the basis for the matrix representation used in the numerical analysis within the following sections in which the Hamiltonian matrix is calculated explicitly. Errors in the numerically obtained results are expressed as relative errors as a percentage of the exact values according to

$$\left| \frac{E_{num} - E_{exact}}{E_{exact}} \right| \times 100\% \quad (4)$$

² see process in Appendix A

3. DIGITISATION

Digitisation refers to the reduction of a continuous scalar field theory, defined over an infinite-dimensional Hilbert space, to a finite representation suitable for implementation on a quantum computer. This is achieved through a combination of discretisation: sampling at finite intervals, and truncation: restriction to a finite domain. This process introduces systematic errors, the analysis and mitigation of which form a primary focus of this work.

On a quantum computer, the number of available basis states is finite and determined by the number of qubits, n_Q . For a single degree of freedom, this yields a Hilbert space of dimension

$$n_\phi = 2^{n_Q}. \quad (5)$$

In the following section, we examine three approaches to digitisation, following similar justification and steps outlined in Klco and Savage [16]. The first is the JLP method which discretises and truncates the field directly in field space and then its finite difference approximation. The third uses a harmonic oscillator (HO) basis, instead digitising in energy eigenstates. The choice of discretisation parameters is constrained by the Nyquist-Shannon sampling theorem, which provides conditions under which digitisation errors can be made exponentially small.

3.1. JLP Basis

The first method of digitisation we investigate is the JLP basis. This provides a good baseline for digitisation errors due to its widespread use. The JLP method is defined by restricting the field to a finite domain $[-\phi_{max}, \phi_{max}]$ with a sampling grid of uniform spacing (δ_ϕ) given by

$$\delta_\phi = \frac{2\phi_{max}}{n_\phi - 1}. \quad (6)$$

and a size determined by the number of available qubits given by Eq. 5. The field then takes values $\phi_j = -\phi_{max} + \delta_\phi \beta_\phi$ where $\beta_\phi = 0, 1, \dots, n_\phi - 1$. Similarly within conjugate momentum space the domain is restricted by k_{max} and a sampling grid is employed, however these are instead determined and fixed by the Fourier relation between field and conjugate momentum space as:

$$k_{max} = \frac{\pi}{\delta_\phi}, \quad \delta_k = \frac{2\pi}{\delta_\phi n_\phi}. \quad (7)$$

With the conjugate momentum taking values as $k_l = -k_{max} + \beta_k \delta_k$ with $\beta_k = 1, 2, \dots, n_\phi$. Together these parameters determine the digitisation errors.

From section 2 the field and conjugate-momentum operators are represented as matrices in a finite-dimensional Hilbert space. We define $\bar{\phi}$ which is diagonal in the field basis with eigenvalues given by the sampled field values ϕ_j and K which is similarly diagonal in the conjugate-momentum basis with eigenvalues given by the discrete momenta k_l . K is the conjugate-momentum basis representation of the non diagonal $\bar{\Pi}$ matrix in the field basis. Here, j and l label the discretised basis states in field and momentum space respectively.

To exploit this structure we apply a Quantum Fourier Transform (QuFoTr), which performs a unitary change of basis between field and momentum representations. This allows the kinetic term to be evaluated efficiently in the momentum basis, where it is diagonal, before transforming back to the field basis. Explicitly, the QuFoTr is defined as:

$$\mathcal{F} |j\rangle = \frac{1}{\sqrt{n_\phi}} \sum_{l=0}^{n_\phi-1} e^{i \frac{2\pi j \cdot l}{n_\phi}} |l\rangle \quad (8)$$

as found in [16]. This transformation maps discrete field eigenstates to conjugate-momentum eigenstates and is the finite dimensional analogue of the continuous Fourier transform relating the field and conjugate-momentum representations. Under this transformation, the momentum operator in the field basis is given by:

$$\bar{\Pi} = \mathcal{F}^\dagger K \mathcal{F}. \quad (9)$$

Because the kinetic term is diagonal in the momentum basis and the potential term within the field basis the full Hamiltonian is not diagonal in either basis.³

The use of a QuFoTr avoids the need for a finite-difference approximation, which otherwise introduces polynomial discretisation errors, instead the discretisation errors introduced are exponentially suppressed once the Nyquist-Shannon sampling conditions are satisfied which we see in the following section 3.1.1. To highlight the advantage of this approach, it is instructive to compare it with this finite-difference approximation which is commonly used in classical lattice formulations. The matrix of the kinetic term can be given by the expression:

$$\bar{\Pi}_{n,m}^2 = \langle \psi_n | \pi^2 | \psi_m \rangle \quad (10)$$

defining π as our conjugate momentum operator $\pi = -i \frac{\partial}{\partial \phi}$, $\pi^2 = -\frac{\partial^2}{\partial \phi^2}$ we see that $\bar{\Pi}$ involves the second order derivative of the wave function $\psi(\phi)$. Using the Taylor expansion of the wave function, the second order derivative is approximated by a finite difference:

$$\psi''(\phi) = \frac{\psi(\phi - \delta\phi) + \psi(\phi + \delta\phi) - 2\psi(\phi)}{\delta_\phi^2} \quad (11)$$

³ For clarity, an example of an obtained Hamiltonian matrix for the JLP method can be found in Appendix B, similar are obtained for all digitisation methods used in this work

we can then write

$$\begin{aligned}
 \bar{\Pi}_{n,m}^2 &= \langle \psi_n(\phi) | -\psi_m''(\phi) \rangle \\
 &= \frac{1}{\delta_\phi^2} \langle \psi_n(\phi) | 2\psi_m(\phi) - \psi_m(\phi - \delta\phi) - \psi_m(\phi + \delta\phi) \rangle \\
 &= \frac{1}{\delta_\phi^2} (2\delta_{n,m} - \delta_{n,m+1} - \delta_{n,m-1}) \\
 &= \frac{1}{\delta_\phi^2} \begin{pmatrix} 2 & -1 & 0 & 0 & \cdots & -1 \\ -1 & 2 & -1 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & -1 & 2 & -1 \\ -1 & 0 & 0 & \cdots & -1 & 2 \end{pmatrix} \tag{12}
 \end{aligned}$$

The form assumes periodic boundary conditions, for which the wave function satisfies $\psi(\phi_{max}) = \psi(-\phi_{max})$. This leads to a wrap around structure within the matrix and is required for consistency with the discrete Fourier transform used in the JLP method where the field and conjugate momentum basis are related through a finite, periodic lattice. Higher order corrections can be introduced to reduce the polynomial error of the finite-difference method. For example the leading correction to the kinetic term is given by $\delta\bar{H} = \frac{1}{24}\delta_\phi^2\bar{\Pi}^4$ derived explicitly in Klcó and Savage [16]. Inclusion of this term and the above matrix as $\bar{\Pi}$ we refer to as the Finite-Difference Correct (FDC) method.

3.1.1. Nyquist Shannon

The significance of the Nyquist-Shannon sampling theorem [18] has been demonstrated to be important when considering quantum simulations such as within [22, 23]. We demonstrate its utility through an example using the JLP method in section 3.1 for a simple harmonic oscillator.

Within the JLP method for our 0+1D Scalar Field Theory we collect information about $\bar{\Pi}$ and $\bar{\phi}$ through a sampling grid, however the resolution and truncation of these two sampling grids are connected by a discrete Fourier transform, leading to reciprocal constraints between the truncations and resolutions within each of the two representations. That is to say as truncation in the field space increases, truncation in the conjugate momentum space decreases, a similar relation is true of the sampling spacing which can be seen through Eq. 6 and Eq. 7.

Within the context of a quantum simulation this implies that both ϕ_{max} and k_{max} must be chosen such that the wave function has negligible support at the boundary of the truncated domains. The Nyquist-Shannon sampling theorem states that when this condition is satisfied, the digitised representation reproduces the continuum wave function with only exponentially small errors. For the case of the ground state wave function of a simple harmonic oscillator this can be seen through simple analysis of the wave function which scales as $\psi_0(\phi) \propto e^{-\frac{\phi^2}{2}}$ giving the probability weight of the truncated region of the field by

$$\int_{|\phi| > \phi_{max}} |\psi_0(\phi)|^2 d\phi \sim e^{-\phi_{max}^2}. \tag{13}$$

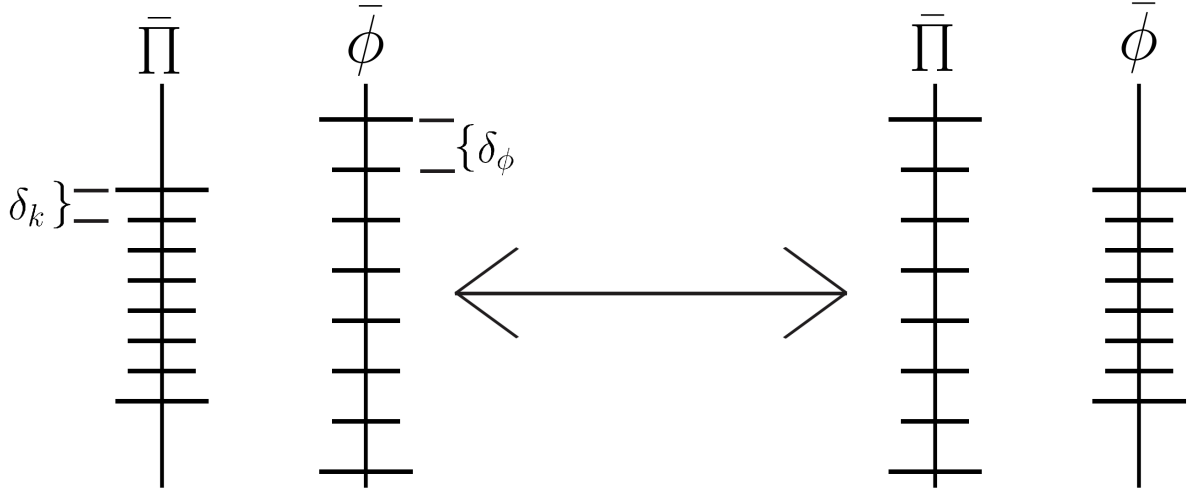


FIG. 1: Schematic illustration of the Nyquist-Shannon sampling constraint in JLP digitisation. The field $\bar{\phi}$ and $\bar{\Pi}$ are represented on a discrete grid with spacings δ_ϕ and δ_k related through a discrete Fourier transform.

This reflects the Gaussian decay of the harmonic oscillator ground state and therefore for sufficiently large truncation the error introduced is exponentially suppressed.

If the condition is violated, aliasing effects lead to significant systematic errors due to under resolution of momentum or field modes. This trade off is illustrated by Figure 1 where increasing truncation in field space reduces the accessible range of conjugate momentum and vice versa. In addition the increase in truncation range leads to decreased detail of the sampled space. The optimal choice therefore balances these so that the physically relevant support of the wave function is captured in both representations whilst maintaining a significant resolution to capture important details. Consequently, once the Nyquist-Shannon conditions are satisfied, digitisation errors become exponentially suppressed with increasing Hilbert space dimension or equivalently the number of qubits within the quantum system.

3.1.2. Deriving a simple Optimal ϕ_{max}

By analogy with the Nyquist-Shannon sampling theorem, an optimal choice of ϕ_{max} can be obtained by balancing resolution in field and conjugate momentum space. For the ground state of a harmonic oscillator, the wave functions in both representations are Gaussian, with standard deviations of σ_ϕ and σ_k respectively. The discretisation spacings δ_ϕ and δ_k determine how well these distributions are sampled. A ratio of the spacing to the deviation provides a measure of sampling resolution to width of distribution, by requiring a comparable sampling of the support in both spaces we can impose

$$\frac{\delta_\phi}{\sigma_\phi} = \frac{\delta_k}{\sigma_k}. \quad (14)$$

For a discretised scalar field, each mode behaves as an independent harmonic oscillator with Hamiltonian

$$H_k = \frac{1}{2}\pi_k^2 + \frac{1}{2}\omega_k^2\phi_k^2, \quad (15)$$

where $\omega_k = \sqrt{k^2 + m^2}$ and k and m are the momentum mode and mass of the field respectively. Working in dimensionless units set by the oscillator frequency the ground state variances are therefore

$$\sigma_\phi^2 = \frac{1}{2\omega_k}, \quad \sigma_k^2 = \frac{\omega_k}{2}. \quad (16)$$

Substituting equations of δ_ϕ and δ_k from Eq. 6 and Eq. 7 and their appropriate variances from Eq. 16 into Eq. 14 we obtain

$$\phi_{max} = \sqrt{\frac{\pi(n_\phi - 1)^2}{2n_\phi}} \quad (17)$$

A similar relation to this can be found in [24], where discretisation is analysed in terms of UV and IR cut-offs arising from a finite lattice and Hilbert space. Within that work optimal parameter choices are inferred from a range of energy scales that can be faithfully represented. This reflects the same underlying need to balance resolution in conjugate variables. The method here makes this balance explicit by applying Nyquist-Shannon considerations directly to the HO ground state. As model complexity increases, the corresponding derivation becomes more involved. The purpose of this example is not generality but to demonstrate that Nyquist-Shannon considerations provide a framework for choosing discretisation parameters in digitised systems, as reflected in the following results. As a primary aim of this work is to minimise computational cost, the ability to determine optimal discretisation parameters a priori, rather than through trial and error tuning, is particularly important.

3.2. Errors of JLP Basis and FDC

Figure 2 shows that for both JLP and FDC a rapid increase in relative error occurs at small ϕ_{max} caused by a truncation of the field that is too small to capture regions where the wave function has appreciable support. We additionally see an increase in relative error at large ϕ_{max} caused by finite grid spacing at a fixed n_Q corresponding to insufficient resolution to resolve important structure within the relevant space.

We see that the FDC method, as expected, has significantly larger errors than those produced by the JLP method. We see that as n_Q increases this disparity increases, with the largest difference in minimum error at $n_Q = 5$ spanning 11 orders of magnitude. This difference can be understood by considering the dominant error sources. The FDC method introduces discretisation errors that scale polynomially in δ_ϕ leading to saturation in achievable accuracy at fixed n_Q . The JLP method employs a QuFoTr to represent the conjugate momentum instead of a finite difference approximation. Therefore once ϕ_{max} is sufficiently large and there is support within the physically relevant regions, digitisation errors become exponentially suppressed due to the Nyquist-Shannon conditions being satisfied. The balance between truncation and discretisation errors in field and conjugate momentum space produces the observed minima consistent with the predictions of Eq. 17 in section 3.1.2. We see with the JLP method rapidly decreasing minimum errors with increasing n_Q at larger ϕ_{max} although with diminishing returns, all of which is consistent with the Nyquist-Shannon sampling theorem.

These JLP results are reproductions of those seen in Klco and Savage [16] and therefore agree

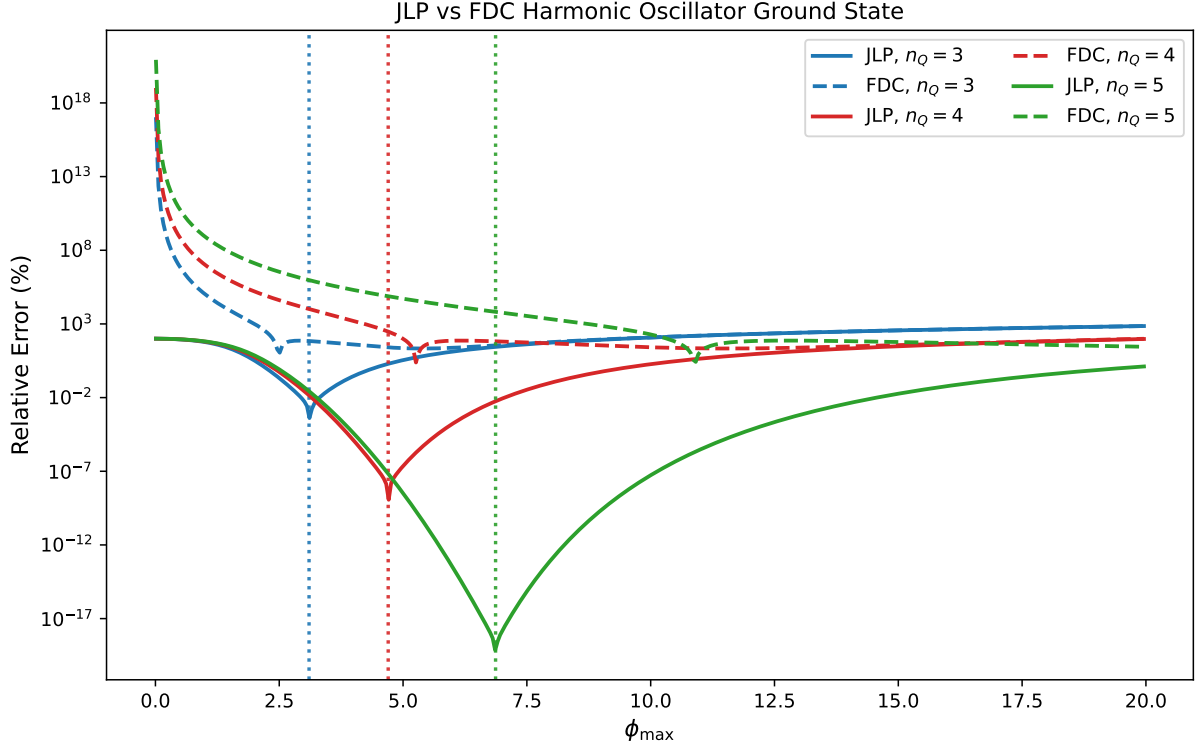


FIG. 2: The expected relative error due to digitisation in the ground state energy ($n = 0$) of the harmonic oscillator defined in Eq. 2, computed using the JLP and FDC methods on an ideal quantum computer for varying n_Q as a function of ϕ_{max} . The reference continuum energy is calculated as $E_n = n + \frac{1}{2}$. Vertical dotted lines indicate the predicted optimal values of ϕ_{max} for JLP for each n_Q given by Eq. 17 .

with the content produced there. These results isolate only the errors from the finite, truncated qubit representation of the field and conjugate momentum coverage and do not include additional error sources such as Trotterisation or hardware noise.

3.3. Harmonic Oscillator (HO) Basis

Whereas the JLP method digitises the field by discretising and truncating its values in field and momentum space, the HO basis uses a truncated basis of harmonic oscillator energy eigenstates. That is, the Hilbert space is approximated by:

$$\{|0\rangle, |1\rangle, \dots, |n_\phi - 1\rangle\} \quad (18)$$

where $|n\rangle$ are eigenstates of the harmonic oscillator Hamiltonian. This approach instead truncates the infinite dimensional Hilbert space to a finite-dimensional subspace spanning low energy states. The field $\bar{\phi}$ and conjugate momentum $\bar{\Pi}$ are constructed using ladder operators of the harmonic oscillator, defined through:

$$H_{HO} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\omega^2\bar{\phi}^2 = \omega(a^\dagger a + \frac{1}{2}). \quad (19)$$

The operators then take the form

$$\bar{\phi} = \frac{a + a^\dagger}{\sqrt{2\omega}} \quad \bar{\Pi} = i\sqrt{\frac{\omega}{2}}(a^\dagger - a) \quad (20)$$

where the ladder operators acting on the basis states are defined as:

$$a|n\rangle = \sqrt{n}|n-1\rangle \quad a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle. \quad (21)$$

This basis is well suited to systems with wave functions localised in field space, such as the harmonic oscillator, as the low energy eigenstates are an efficient representation of the physically relevant regions. Consequently, for such systems, accurate results can often be obtained with fewer basis states compared to direct discretisation in field space. Within this basis, the operators $\bar{\phi}$ and $\bar{\Pi}$ take dense matrix forms in contrast to the diagonal structure of operators in the JLP field basis.

The oscillator frequency ω acts as the tuning variable within this regime and therefore must be chosen appropriately. When tuned to match the typical curvature of the wave function, low energy states can be captured efficiently while contributions from higher energy states are suppressed by the finite basis size. Although the HO basis does not impose a hard cut-off in field space, truncation of the basis means there is an effective limitation on the accessible field space support and high momentum components. With regards to the Nyquist-Shannon sampling theorem, the HO basis reduces the number of required basis states by matching the localisation properties of the low energy wave functions in both field and conjugate momentum space, rather than enforcing uniform resolution across the entire field domain. For states that are not well localised in field space or that exhibit support at large field values, the HO basis may require a large number of basis states to achieve a comparable accuracy to the JLP method.

3.4. Errors of JLP and HO Basis

An anharmonic oscillator provides a non-trivial test of the HO basis. As the basis is constructed from the harmonic oscillator eigenstates, a purely harmonic Hamiltonian would be represented exactly apart from truncation and would therefore obscure true performance of the basis and its digitisation errors. The structure of the JLP and HO basis anharmonic relative errors within Figure 3 follow a similar overall structure to those found in the harmonic case with respect to their tuning parameter. In each case the curve exhibits a clear minimum. As before, within the JLP method we see high errors at small ϕ_{max} due to insufficient sampling of significant structure and conversely at large ϕ_{max} additional high errors due to limited sampling resolution. A minimum is found when these effects balance in both field and conjugate-momentum space as per the Nyquist-Shannon sampling theorem. A similar structure is present for the HO basis errors. At small ω , basis states are overly delocalised leading to poor resolution of the ground state. Whereas at large ω , the basis is overly localised and its ability to accurately represent the support of the ground state wave function is reduced. Within Figure 3 we also see small non-monotonic bumps, these are attributed to finite-basis truncation effects. The variation of the tuning parameter leads to the overlap of the exact ground state with the truncated basis to change non-smoothly, this leads to local oscillations in the relative error.

A similar trend is also seen as in the previous analysis with respect to the qubit number. As n_Q increases from 3 to 5 we see a decrease in minimum digitisation error across both the HO basis and the JLP basis, as expected due to the increased dimensionality of the truncated Hilbert

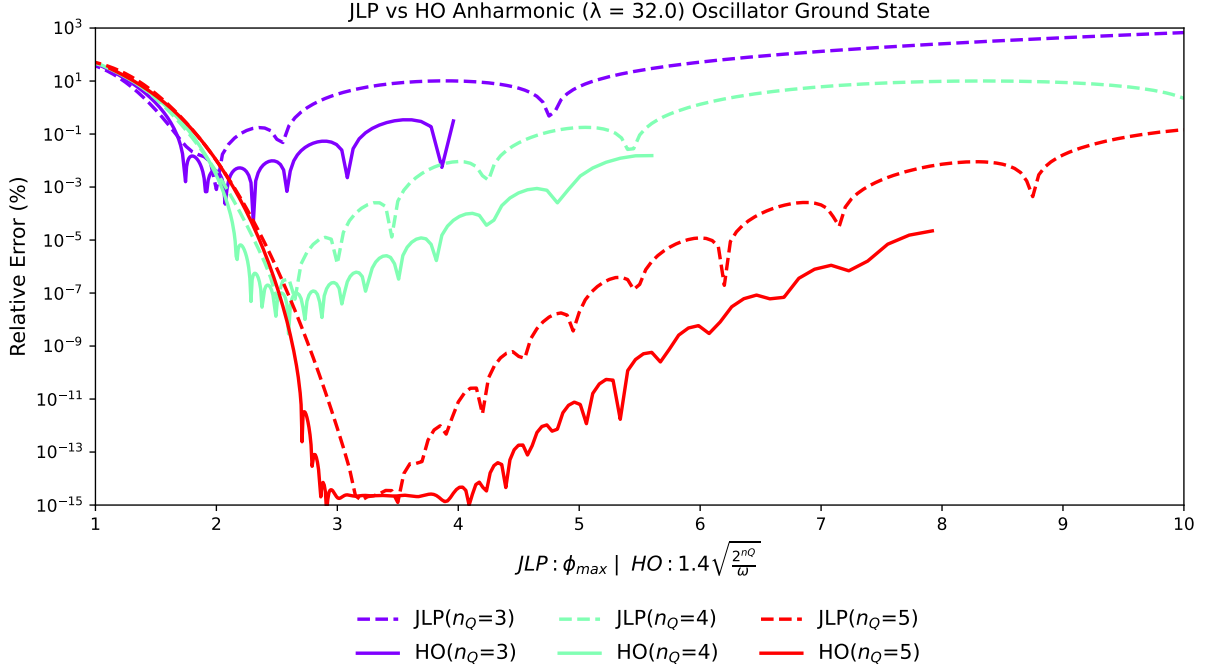


FIG. 3: The expected relative error due to digitisation in the ground state energy ($n = 0$) of the anharmonic oscillator defined in Eq. 3 with $\lambda_0 = 32$, computed using the JLP and HO basis on an ideal quantum computer for varying n_Q as a function of ϕ_{max} and ω respectively with a scaled horizontal axis of $1.4\sqrt{\frac{2^{n_Q}}{\omega}}$ for the HO basis. The reference continuum energy is obtained using the method described in Appendix A. .

space. We see that for all values of n_Q the HO basis produces lower minimum errors, however as n_Q increases this difference becomes smaller. This indicates that for the low energy anharmonic ground state the HO basis provides a more efficient description. This reflects the fact that the HO basis is variationally adapted to low energy states: the basis functions are already localised to the region where the ground state has significant support. Therefore this basis is better matched to the wave function than the uniformly distributed basis used in the JLP method. The difference between the minimum errors of these methods becoming smaller is consistent with both approaches converging toward the exact continuum result.

Within Figure 3 the HO horizontal axis is plotted in terms of $1.4\sqrt{\frac{2^{n_Q}}{\omega}}$. This relation arises from the harmonic oscillator itself, using the characteristic width of the quantum harmonic oscillator eigenfunctions we obtain

$$\sigma_\phi = \sqrt{\langle \phi^2 \rangle} \sim \sqrt{\frac{n}{\omega}}. \quad (22)$$

For a truncated basis $n = n_\phi = 2^{n_Q}$, $\sqrt{\frac{2^{n_Q}}{\omega}}$ then plays a role analogous to ϕ_{max} in the JLP basis. The numerical factor of 1.4 is an empirical rescaling introduced only to align the two parametrisation on the same plot for clearer comparison.

These results are a reproduction of those seen in Klco and Savage [16] and thus for direct comparison their used value of λ_0 is employed. We find good agreement with [16]. Although the JLP method is more commonly used, the HO basis yields reduced digitisation errors in

the case of the produced anharmonic oscillator simulations. In addition the HO basis is more suitable for use in section 4 as it generalises naturally to be used in Hamiltonian Truncation Effective Theory. Again these results isolate only the errors from the finite, truncated qubit representation of the field and conjugate momentum coverage and do not include additional error sources such as Trotterisation or hardware noise.

4. HAMILTONIAN TRUNCATION EFFECTIVE THEORY

The digitisation methods discussed in the previous section reduce the infinite dimensional Hilbert space of scalar field theories to a finite representation which is suitable for computation on quantum devices. This is necessary but introduces truncation errors. It is then a natural question to ask if these errors can be reduced systematically without the need for increasing the Hilbert space dimension, and therefore the required qubit cost.

Cohen *et al.* [25] introduces Hamiltonian truncation effective theory (HTET) which provides a potential avenue for this purpose. Instead of treating the truncation as a purely numerical approximation, HTET treats the energy cut-off E_{max} , which is tied to the truncation of the basis and therefore the number of qubits, as a physical ultraviolet cut-off and then seeks to incorporate the effects of the excluded high energy states by use of an effective Hamiltonian acting within the truncated space. This effective Hamiltonian is organised as an expansion of inverse powers of E_{max} , which attempt to reduce truncation errors order by order. HTET aims to improve the accuracy of low energy observables while keeping the computational basis we use fixed and is therefore well suited to the present context.

In this section, we introduce the formalism of HTET specialised to the HO basis introduced in section 3.3 and assess its effectiveness in reducing digitisation errors through numerical study. We begin by outlining the general construction of the Hamiltonian, deriving the correction terms used in this work as seen in Cohen *et al.* [25], discuss issues of non-Hermiticity and finally present numerical results as functions of the basis frequency ω and truncation energy E_{max} .

4.1. The Effective Hamiltonian

To apply HTET within the 0+1 dimensional setting we work in the HO basis introduced in section 3.3. We follow the general construction seen in Cohen *et al.* [25]. The full Hamiltonian is decomposed into a solvable part and an interaction term

$$\bar{H} = \bar{H}_0 + \bar{V}. \quad (23)$$

The low energy effective Hilbert space is defined by the eigenstates of $\bar{H}_0$ whose energies lie below a cut-off E_{max} . The goal of HTET is to construct an effective Hamiltonian that acts only within the truncated space while systematically incorporating the effects of the excluded states through perturbative matching.

We consider the dimensionless anharmonic oscillator hamiltonian introduced in section 2,

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2 + \frac{\lambda_0}{4!}\bar{\phi}^4. \quad (24)$$

HTET requires a decomposition into a reference Hamiltonian $\bar{H}_0$ and an interaction $\bar{V}$. Introducing an auxiliary frequency, we rewrite

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\omega^2\bar{\phi}^2 + \frac{1}{2}(1 - \omega^2)\bar{\phi}^2 + \frac{\lambda_0}{4!}\bar{\phi}^4 \quad (25)$$

defining

$$\bar{H}_0 = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\omega^2\bar{\phi}^2 \quad \bar{V} = \frac{1}{2}(1 - \omega^2)\bar{\phi}^2 + \frac{\lambda_0}{4!}\bar{\phi}^4 \quad (26)$$

The parameter ω is not physical. It determines the spacing and structure of the HO basis used to define $\bar{H}_0$. An E_{max} sets the cut-off on how many of these states are retained and has units of dimensionless energy. This is analogous to the Hilbert space truncation in the digitised field basis, where the dimension is fixed by Eq. 5 as $n_\phi = 2^{n_Q}$. To the same effect then at finite E_{max} observables exhibit an ω dependence which vanishes as $E_{max} \rightarrow \infty$.

The operators $\bar{\phi}$ and $\bar{\Pi}$ introduced by the HO basis in section 3.3 and the appropriate ladder operators remain valid and similarly the eigenstates of $\bar{H}_0$ define the working basis,

$$\bar{H}_0 |n\rangle = E_n^{(0)} |n\rangle, \quad E_n^{(0)} = \omega(n + \frac{1}{2}) \quad (27)$$

with the truncated Hilbert space defined by retaining only states satisfying

$$E_n^{(0)} \leq E_{max} \quad (28)$$

An effective Hamiltonian H_{eff} is to be constructed that acts within this space,

$$\bar{H}_{eff} = \bar{H}_0 + H_1 + H_2 + H_3 + H_4 + \dots, \quad H_n = O(\bar{V}^n) \quad (29)$$

however matching eigenvalues alone does not uniquely determine $\bar{H}_{eff}$ as shown in Cohen *et al.* [25] as perturbative eigenvalues, E_n depend on off diagonal matrix elements of lower order operators as well as the diagonal matrix elements of higher order terms. Instead the effective Hamiltonian must be fixed by matching transition amplitudes through a transition matrix.

To define these amplitudes, we introduce an adiabatic regulator

$$\bar{V} \rightarrow \bar{V} e^{-\epsilon t}. \quad (30)$$

for $t > 0$ with an initial state at $t_i = 0$ and defining ϵ as an infinitesimal regulating parameter with units of energy where $\epsilon > 0$. This ensures convergence of time integrals. Attempting to define the Hamiltonian by matching time evolution amplitudes between the low energy states results, in the limit $t_f \rightarrow \infty$, these amplitudes to contain an ill defined infinite phase terms. Instead we work in the interaction picture,

$$|\Psi(t)\rangle_{IP} = e^{i\bar{H}_0 t} |\Psi(t)\rangle_{SP}, \quad (31)$$

by differentiating with respect to time and inserting the Schrödinger equation with the decomposition of the Hamiltonian Eq. 23, the regulated interaction term Eq. 30 and $|\Psi(t)\rangle_{SP}$ Eq. 31

we obtain

$$i \frac{d}{dt} |\Psi\rangle_{IP} = e^{i\bar{H}_0 t} \bar{V} e^{-\epsilon t} e^{-i\bar{H}_0 t} |\Psi\rangle_{IP} = V_{IP}(t) |\Psi(t)\rangle_{IP} \quad (32)$$

Defining the interaction picture evolution operator U_{IP} by

$$|\Psi(t)\rangle_{IP} = U_{IP}(t, t_i) |\Psi(t_i)\rangle_{IP} \quad (33)$$

which satisfies

$$\frac{\partial}{\partial t_i} U_{IP}(t_f, t_i) = i U_{IP}(t_f, t_i) V_{IP}(t_i), \quad \text{with} \quad U_{IP}(t_f, t_f) = \mathbf{1} \quad (34)$$

integrating from t_i to t_f ,

$$U_{IP}(t_f, t_i) = \mathbf{1} - i \int_{t_i}^{t_f} dt U_{IP}(t_f, t) V_{IP}(t) \quad (35)$$

we obtain a formal solution which is a type of Volterra integral. Forming this as an iterative solution leads to a Dyson series [26] and the following path ordered exponential is obtained

$$U_{IP}(t_f, t_i) = \mathcal{T} \exp\left\{-i \int_{t_i}^{t_f} dt V_{IP}(t)\right\}. \quad (36)$$

V_{IP} is time dependent and does not generally commute at different times and $\mathcal{T}$ is the time ordering operator. Defining the operator ξ^4 , which plays the role of a regulated analogue of the S-matrix appropriate to the truncated Hilbert space, where $\epsilon \neq 0$:

$$\langle f | \xi(\epsilon) | i \rangle = \lim_{t_f \rightarrow \infty} \langle f | U_{IP}(t_f, 0) | i \rangle. \quad (37)$$

which describes the transition from an initial state $|i\rangle$ to a final state $|f\rangle$. Re-expanding U_{IP} generates a perturbative expansion in powers of $\bar{V}$. At zeroth order

$$\begin{aligned} \langle f | \xi^{(0)} | i \rangle &= \langle f | U_{IP}(t_i, 0) | i \rangle \\ &= \delta_{fi}. \end{aligned} \quad (38)$$

At first order

$$\langle f | \xi^{(1)} | i \rangle = (-i) \int_0^\infty dt \langle f | e^{i\bar{H}_0 t} \bar{V} e^{-\epsilon t} e^{-i\bar{H}_0 t} | i \rangle. \quad (39)$$

Given that $|i\rangle$ and $|f\rangle$ are eigenstates of $\bar{H}_0$ and time evolution produces a phase of $e^{i(E_f - E_i)t}$, we obtain

$$\langle f | \xi^{(1)} | i \rangle = (-i) \langle f | \bar{V} | i \rangle \int_0^\infty dt e^{i(E_f - E_i - \epsilon)t} = \frac{\langle f | \bar{V} | i \rangle}{E_f - E_i + i\epsilon}. \quad (40)$$

Repeating this procedure yields

$$\langle f | \xi | i \rangle = \delta_{fi} + \frac{\langle f | \bar{V} | i \rangle}{E_{fi} + i\epsilon} + \sum_\alpha \frac{\langle f | \bar{V} | \alpha \rangle \langle \alpha | \bar{V} | i \rangle}{(E_{fi} + i\epsilon)(E_{f\alpha} + i\epsilon)} + O(\bar{V}^3), \quad (41)$$

where $E_{f\alpha} = E_f - E_\alpha$. It is convenient to isolate the interaction dependent part of the amplitude by defining the T matrix

$$\langle f | \xi | i \rangle = \delta_{fi} + \frac{\langle f | T | i \rangle}{E_{fi} + i\epsilon}. \quad (42)$$

⁴ we use the symbol ξ for clarity here instead of the $\sum$ symbol used in Cohen *et al.* [25] as $\sum$ are used consistently for summation within this derivation

The perturbative expansion of T can then be given by

$$\langle f | T_n | i \rangle = \sum_{\alpha_1, \dots, \alpha_{n-1}} \frac{\langle f | \bar{V} | \alpha_1 \rangle \langle \alpha_1 | \bar{V} | \alpha_2 \rangle \cdots \langle \alpha_{n-1} | \bar{V} | i \rangle}{(E_{f\alpha_1} + i\epsilon) \cdots (E_{f\alpha_{n-1}} + i\epsilon)}, \quad (43)$$

with the leading order terms written as

$$\langle f | T | i \rangle = \langle f | \bar{V} | i \rangle + \sum_{\alpha} \frac{\langle f | \bar{V} | \alpha \rangle \langle \alpha | \bar{V} | i \rangle}{E_{f\alpha} + i\epsilon} + \sum_{\alpha, \beta} \frac{\langle f | \bar{V} | \alpha \rangle \langle \alpha | \bar{V} | \beta \rangle \langle \beta | \bar{V} | i \rangle}{(E_{f\alpha} + i\epsilon)(E_{f\beta} + i\epsilon)} + O(\bar{V}^4). \quad (44)$$

This expansion contains terms whose denominators vanish as $\epsilon \rightarrow 0$, so the transition matrix is formally infrared divergent before matching. Following Cohen *et al.* [25], ϵ is treated as an IR regulator and these divergences cancel in the matching to the effective theory. The effective Hamiltonian is defined such that it reproduces these transition amplitudes within the truncated Hilbert space order by order in $\bar{V}$. Writing

$$V_{eff} = H_1 + H_2 + H_3 + \cdots, \quad (45)$$

and restricting the intermediate states appropriately by $E \leq E_{max}$. The effective Hamiltonian is defined such that transition amplitudes within the low energy subspace match those of the full theory order by order,

$$\langle f | H_1 | i \rangle_{eff} = \langle f | \bar{V} | i \rangle \quad (46a)$$

$$\langle f | H_2 | i \rangle_{eff} = \sum_{\alpha}^> \frac{\langle f | \bar{V} | \alpha \rangle \langle \alpha | \bar{V} | i \rangle}{E_{f\alpha}} \quad (46b)$$

$$\langle f | H_3 | i \rangle_{eff} = \sum_{\alpha, \beta}^> \frac{\langle f | \bar{V} | \alpha \rangle \langle \alpha | \bar{V} | \beta \rangle \langle \beta | \bar{V} | i \rangle}{E_{f\alpha} E_{f\beta}} - \sum_{\alpha}^< \sum_{\beta}^> \frac{\langle f | \bar{V} | \alpha \rangle \langle \alpha | \bar{V} | \beta \rangle \langle \beta | \bar{V} | i \rangle}{E_{\alpha\beta} E_{f\beta}}. \quad (46c)$$

Here $\sum_{\alpha}^< \equiv \sum_{E_{\alpha}^{(0)} \leq E_{max}}$ and $\sum_{\alpha}^> \equiv \sum_{E_{\alpha}^{(0)} > E_{max}}$ and the subscript *eff* indicates that the left hand matrix is evaluated within the truncated effective Hilbert space. At second order in Eq. 46b the virtual excursions into the high energy are encoded. The system begins in a low energy state $|i\rangle$, it is acted upon by $\bar{V}$ which connects it to a high energy state $|\alpha\rangle$. The system then returns to a low energy state $|f\rangle$ via a second insertion of the potential. A similar process occurs in Eq. 46c. The first term describes a higher order entirely within the high energy sector over three steps. The second term subtracts contributions involving intermediate low energy states ensuring processes already accounted for by lower order terms H_1, H_2 are not double counted.

An important note to make is that the effective Hamiltonian constructed in this manner is not, in general Hermitian at finite order in the expansion. This non-Hermiticity arises from the asymmetric structure of the energy denominators, such as $E_{f\alpha}$, which explicitly depends on the final state and therefore breaks symmetry under interchange of initial and final states. Fundamentally this reflects that the effective Hamiltonian is defined through matching transition amplitudes rather than through a variational or operator based projection. It is therefore not constrained to be Hermitian order by order. Within section 4.1.2 this is discussed and does not lead to inconsistency, the effective Hamiltonian reproduces the correct low energy transition amplitudes and eigenvalues within the truncated space. In addition the non-Hermiticity vanishes as higher order corrections are included.

These expressions demonstrate explicitly how HTET incorporates the effects of virtual transitions into the excluded higher energy sector without increasing Hilbert space dimension. This offers potential improvements to accuracy without increasing qubit requirement for the process of digitisation.

4.1.1. Assumptions

The HTET construction makes several key assumptions the most relevant of which for the present application are:

1. The full Hamiltonian admits a decomposition $\bar{H} = \bar{H}_0 + \bar{V}$, where $\bar{H}_0$ is exactly diagonalizable with a known discrete spectrum, providing a well defined basis for truncation
2. The interaction $\bar{V}$ is perturbative within the ultraviolet, such that contributions from states above the cut-off can be incorporated through a controlled expansion in powers of $\bar{V}$
3. The low energy effective Hilbert space is defined by the states satisfying $E_n^{(0)} \leq E_{max}$ and the matching corrections are assumed to exhibit separation of scales within the context of effective field theories: they are dominated by states near the cut-off and can therefore be organised as corrections suppressed by inverse powers of E_{max} .

Under these assumptions, the effects of the truncated high energy sector can be incorporated perturbatively and an effective Hamiltonian can provide an approximation to the low energy spectrum. When these assumptions cease to hold clearly, or when the finite order expansion fails to remain well ordered, the HTET approximation can lose predictive control, potentially producing regions of instability and large error.

4.1.2. Non-Hermiticity

In addition to the assumptions outlined above, a notable feature of HTET is that the effective Hamiltonian, in general, defined by this procedure is non-Hermitian at finite order. As emphasised in Cohen *et al.* [25] that this does not arise from an inconsistency within the underlying theory but instead from the manner in which the effective Hamiltonian is defined.

Within HTET, H_{eff} is fixed by matching the transition matrix T . We require that matrix elements of T in the truncated Hilbert space reproduce those of the full theory order by order within perturbation theory. The matching conditions express the effective Hamiltonian matrix elements as sums over intermediate states with energy denominators of the form $E_{f\alpha} = E_f - E_\alpha$. These denominators depend explicitly on the final state and therefore the resulting matrix elements are not symmetric under the interchange of the initial and final states. Consequently, we do not generally have

$$\langle f | H_{eff} | i \rangle = \langle i | H_{eff} | f \rangle^* \quad (47)$$

order by order in the expansion.

The lack of Hermiticity reflects that the full Hamiltonian couples the retained low energy subspace to high energy states that have been integrated out. As a result, time evolution with the full Hamiltonian generically mixes states within the truncated space with those in the excluded space such that the truncated Hilbert space is not invariant under this evolution. The effective Hamiltonian then is not required to generate a closed, unitary time evolution within truncated space, but instead to reproduce the matched transition amplitudes of the full theory. Consequently there is no requirement that the finite dimensional operator H_{eff} be Hermitian [25].

It is natural to then ask whether an alternative, explicitly Hermitian definition of the effective Hamiltonian can be constructed. However attempts to impose Hermiticity directly can spoil the effective field theory structure. By defining a Hermitian operator by matching $T + T^\dagger$ we would yield an effective Hamiltonian that develops infrared divergences at higher order in perturbation theory. These divergences arise from vanishing energy denominators and signal the breakdown of the separation of scales that underpins the construction of HTET.

The non-Hermiticity of H_{eff} at finite order is a controlled and well understood feature of the formalism. This is not problematic, since H_{eff} is not required to generate a unitary time evolution within the truncated space, but only to reproduce the correct low energy transition amplitudes and observables of the full theory, order by order. It is a consequence of matching an inherently asymmetric object, the transition matrix, in a truncated space where high energy degrees of freedom have been systematically integrated out. Importantly the construction ensures that physical observables in the low energy sector are reproduced order by order despite the lack of Hermiticity at intermediate stages.

4.2. Simulation

We aim to obtain numerical results for HTET for the dimensionless 0+1 dimensional scalar field theory. The system is represented in the HO basis and truncated to states with unperturbed energies satisfying $E_n^{(0)} \leq E_{max}$ defining the low energy Hilbert space.

For each choice of (ω, E_{max}) , matrix elements of the interaction $\bar{V}$ were computed between HO basis states and the HTET matching corrections were evaluated using the expression given in Eq. 46. The resulting finite dimensional effective Hamiltonian is then assembled order by order and diagonalised numerically to obtain approximations of the ground state energy of an anharmonic oscillator.

To assess accuracy, the computed energies were compared with independently obtained high precision reference values, described in Appendix A. The relative error is defined as in Eq. 4

$$\left| \frac{E_{num} - E_{exact}}{E_{exact}} \right| \times 100\% \quad (4)$$

$\lambda_0 = 1$ is chosen for testing HTET as the method is formulated for theories in which the interaction is perturbative in the ultraviolet, providing a regime in which the expansion is expected to be well controlled and results may act as a clearer benchmark. The procedure is repeated over a grid of ω from 0.3-2.75 and E_{max} from 1-100. These ranges were chosen to appropriately show

the significant behaviour of the respective models used. All calculations were performed using high precision (30 decimal places) numerical arithmetic such that the observed deviations can be attributed to truncation and HTET approximations rather than floating point error.

4.3. Errors of Hamiltonian Truncation Effective Theory

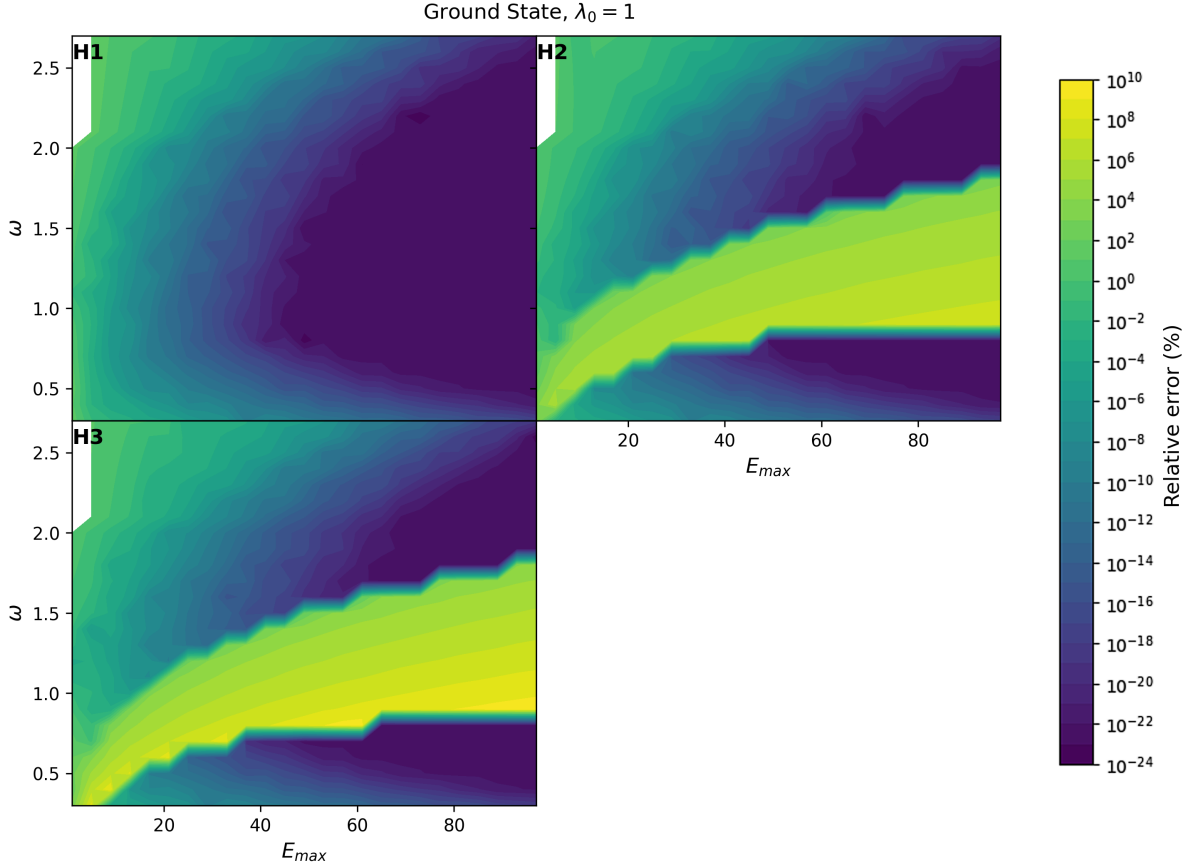


FIG. 4: The expected relative error due to digitisation in the ground state energy ($n = 0$) of the anharmonic oscillator defined in Eq. 3 with $\lambda_0 = 1$, computed using the HO basis on an ideal quantum computer, as a function of ω and E_{max} . The reference continuum energy is obtained using the method described in Appendix A. Panels **H1-H3** show results obtained with increasing orders of the effective Hamiltonian from Eq. 46. **H1** corresponds to the uncorrected HO basis, **H2** includes the first order correction, and **H3** includes both first and second order corrections. White space within these graphs correspond to regions where the truncation contains too few states to define a meaningful effective Hamiltonian.

Figure 4 shows the relative error in the ground state energy as a function of E_{max} and ω for increasing orders of the effective Hamiltonian. As expected, increasing the truncation limit E_{max} leads to reduced digitisation errors across all implementations ($H_1 - H_3$) due to the improved coverage of the Hilbert space. Additionally, a dependence on ω is observed, consistent with the Nyquist-Shannon considerations which constrain how accurately the wave function support is captured [18]. As E_{max} increases, the ω dependence weakens and all methods approach errors of order 10^{-24} . For H_1 , the error varies smoothly across both parameters and exhibits only mild non-monotonic behaviour in both E_{max} and ω . No sharp features are present, and the error is

dominated by the expected truncation effects and basis mismatch between the HO basis and the interacting ground state. In contrast to this stability, H_2 and H_3 exhibit a pronounced band of large errors. Outside this region, the qualitative behaviour resembles that of H_1 however within this band the error increases by substantial orders of magnitude. This indicates a loss of numerical control in the finite order HTET expansion when implemented in the ω dependent HO basis.

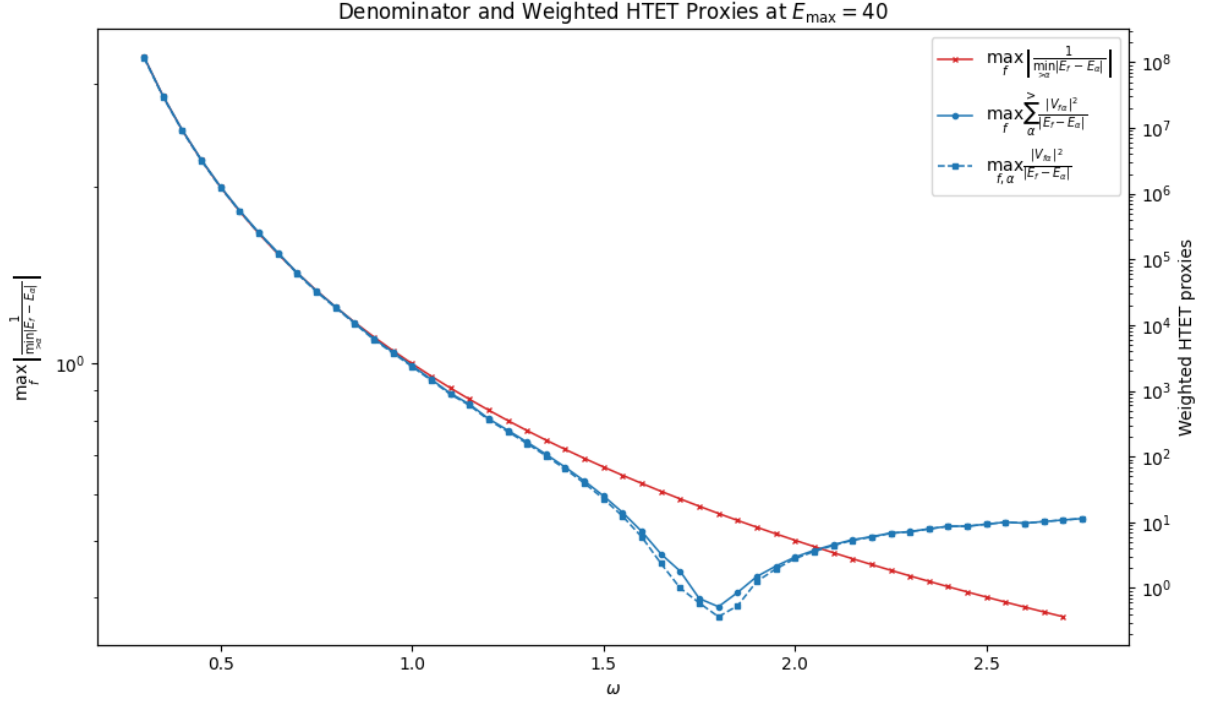


FIG. 5: Magnitudes of proxy equations of components of the HTET corrections at a fixed $E_{max} = 40$ as a function of ω .

The effective Hamiltonians of H_2 and H_3 from Eq. 46 contain energy denominators of the form

$$\frac{1}{E_f - E_\alpha}. \quad (48)$$

With E_f corresponding to energies within the truncated space, whilst E_α corresponds to the excluded states. The validity of the expansion of the effective Hamiltonians relies on the matching corrections being controlled by states near the cut-off, so that the relevant energy denominators remain of order E_{max} and resulting contributions admit an expansion in inverse powers of E_{max} . One could imagine that if $E_f \approx E_\alpha$ these denominators would become extremely small, leading to large contributions from the corresponding terms which would be consistent with a loss of the expected separation of scales. However plotting only the maximum contributions to the second order corrections from denominators in 46b in figure 5 we see no extreme contribution that could naturally explain the instability bands seen in both H_2 and H_3 . The denominators alone do not exhibit singular behaviour, the full HTET corrections also depend

on matrix elements $V_{f\alpha}$ and their summation. Therefore by defining weighted proxies of

$$\max_f \left\{ \sum_{\alpha}^> \frac{|V_{f\alpha}|^2}{E_f - E_{\alpha}} \right\} \quad (49)$$

$$\max_{f\alpha} \left\{ \frac{|V_{f\alpha}|^2}{E_f - E_{\alpha}} \right\} \quad (50)$$

we able to investigate the effects of introducing these matrices on the HTET corrections. These too also exhibit no sharp features in figure 5 that could contribute to the instability band. Instead, all three vary smoothly with ω . This indicates that the instability band is not driven by either small denominators, large matrix elements, or their combination within HTET contributions. As the magnitude of individual HTET contributions are not the cause of the instability it is then a natural step to investigate whether these instability bands are manifested from perturbative control in the HTET expansion.

By defining a quantity

$$\frac{\|H_n - H_{n-1}\|_F}{\|H_1\|_F} \quad (51)$$

where $\|Q\|_F$ is the Frobenius norm of a matrix and H_n is a matrix corresponding to the n^{th} order HTET correction, we are able to obtain the relative size of the successive HTET correction matrices with respect to the uncorrected H_1 ⁵. If the cause of the instability band is introduced through the corrections of HTET losing perturbative control $\|H_n\| \not\ll \|H_{n-1}\|$, we would expect these ratios to become large within the same regions as the instability bands.

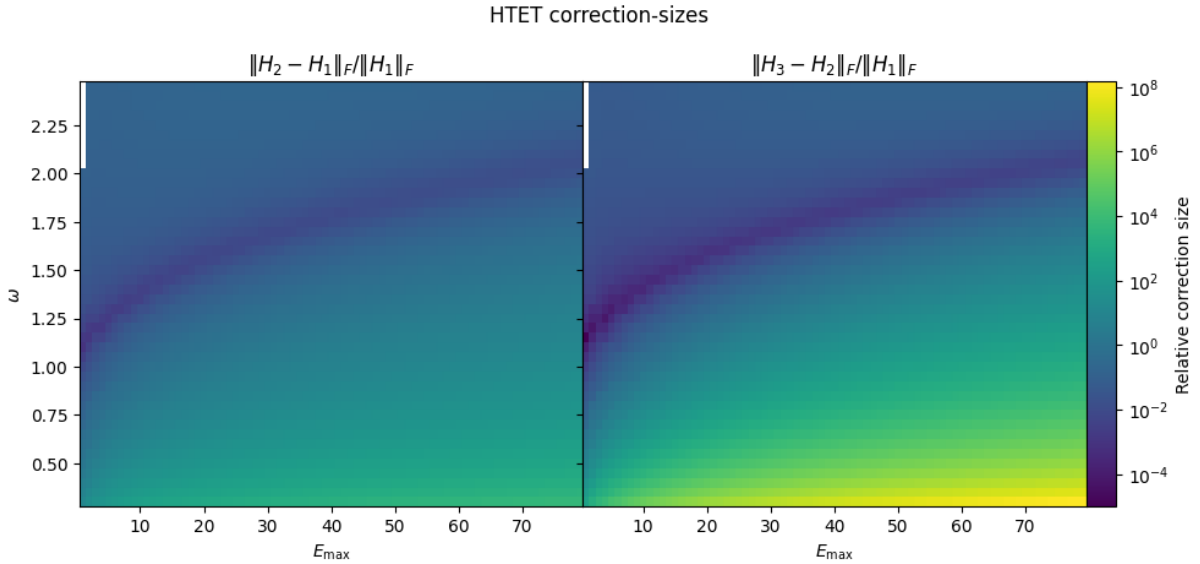


FIG. 6: Relative size of the successive HTET correction matrices as functions of ω and $E_{\max}$ given by Eq. 51. White space within these graphs correspond to regions where the truncation contains too few states to define meaningful Hamiltonian terms.

Within figure 6 we observe distinct regions of high relative difference between contributions

⁵ The Frobenius norm is defined as $\|Q\|_F = \sqrt{\sum_{ij} |Q_{ij}|^2}$, it is used here as it provides a convenient scalar measure of the overall size of the matrix correction and is therefore well suited to diagnosing loss of perturbative control however it does not capture sensitivity of individual eigenvalues to structured changes in the Hamiltonian.

from H_2 and H_3 with H_1 however no distinct high value band is present similar to that within figure 4. This implies that the instability band structure cannot be attributed solely to the overall magnitude of the higher order corrections.

The structure of the instability band does not originate from the growth of matrix sizes or energy denominators, or due to a breakdown in the relative size of successive HTET corrections. Instead then, we suggest that it arises from the dependence of the truncated Hilbert space on the oscillator frequency ω applied to the context of HTET. The set of states satisfying $E_n^{(0)} \leq E_{max}$ is dependent on ω . As ω is varied, harmonic oscillator eigenstates cross the truncation boundary. When such a crossing occurs, a state near the truncation boundary transitions abruptly from being treated as exact, within the truncated region, to being treated perturbatively outside the truncated space through the effective Hamiltonian. This behaviour undermines the practical scale separation required for a controlled HTET expansion, since states near the truncation boundary can switch discontinuously between treatments⁶.

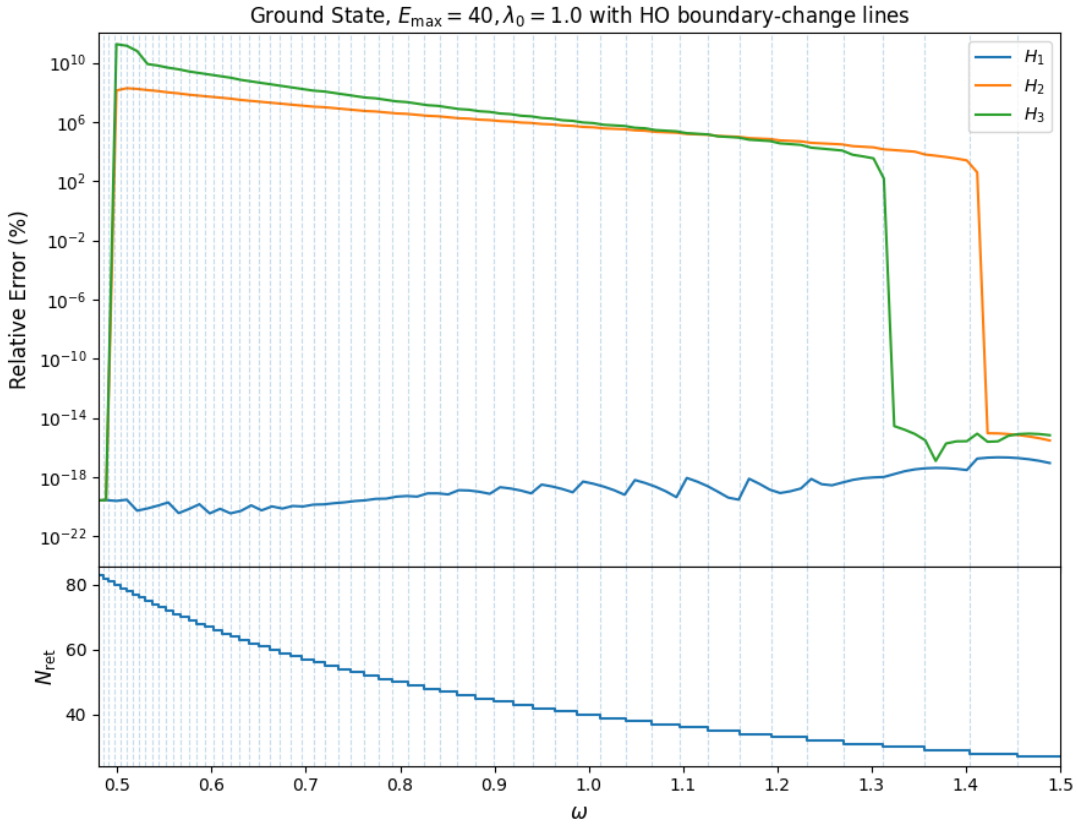


FIG. 7: The expected relative error due to digitisation in the ground state energy ($n = 0$) of the anharmonic oscillator defined in Eq. 3 with $\lambda_0 = 1$, computed using the HO basis on an ideal quantum computer, as a function of ω over a reduced range at a fixed $E_{max} = 40$, chosen as to capture the instability band structure, for correction $H_1 - H_3$ for the effective Hamiltonian. Vertical dashed lines indicate values of ω at which HO eigenstates satisfy $E_n^{(0)} = E_{max}$ corresponding to changes within the retained Hilbert space. The lower panel shows the number of retained states N_{ret} which decreases stepwise with ω .

⁶ This is distinct from the diagnostic based on the individual energy denominators $E_f - E_\alpha$, here the issue is instead the discontinuous reassignment of near boundary states between the retained and excluded sectors as ω varies

This interpretation is supported by figure 7, which shows a slice of constant E_{max} from figure 4, in which a sharp, discontinuous change in the error is observed, which is in alignment with dashed lines marking changes in the retained Hilbert space. These dashed lines arise from the stepwise variation in the number of retained states (N_{ret})

$$N_{ret} = \text{floor} \left[\frac{E_{max}}{\omega} + \frac{1}{2} \right] \quad (52)$$

coming from defining retention of states as $E_n^{(0)} = \omega(n + \frac{1}{2}) \leq E_{max}$ with $N_{ret} = n + 1$. This therefore leads to discrete changes at

$$\omega = \frac{E_{max}}{n + \frac{1}{2}}. \quad (53)$$

Furthermore within 7 we see H_3 exhibits larger errors than H_2 which is consistent with an increased sensitivity to contributions from states across the truncation boundary. A counter intuitive feature is also observed: within the instability region of figure 4, the error increases as E_{max} increases. Whilst increasing the Hilbert space typically improves accuracy, it also increases the density of states near the truncation boundary by nature of more states included within the boundary. This would increase the likelihood of near boundary contributions.

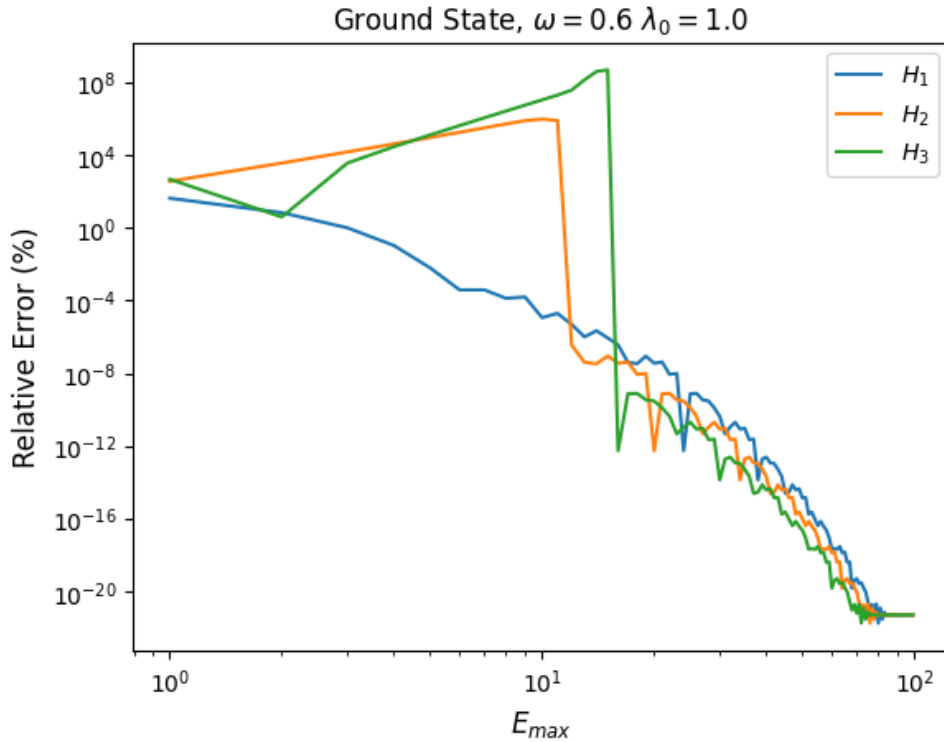


FIG. 8: The expected relative error due to digitisation in the ground state energy ($n = 0$) of the an-harmonic oscillator defined in Eq. 3 with $\lambda_0 = 1$, computed using the HO basis on an ideal quantum computer, as a function of E_{max} for a fixed $\omega = 0.6$, for correction terms $H_1 - H_3$ for the effective Hamiltonian. $\omega = 0.6$ is chosen as it captures both within and outside the instability region .

In addition to observing trends at fixed E_{max} we also see in figure 8 a slice of varying E_{max} at fixed ω . This allows us to clearly analyse the performance of each correction. Outside the instability region H_3 outperforms H_2 and H_2 outperforms H_1 achieving reduced errors at lower

values of E_{max} . This improvement over the uncorrected HO basis is quantified in table I which shows that the same minimum error can be achieved with a reduced E_{max} when higher order corrections are included. This corresponds directly to a reduction in the required Hilbert space dimension and therefore qubit cost. In totality then we identify a restricted region in which HTET provides a controlled improvement over an uncorrected HO basis. Although ω should be chosen such that the truncation yields a practically controlled HTET expansion, with states near the truncation boundary not producing enhanced sensitivity.

	% Min-Error	ω	E_{max}
H_1	4.48×10^{-24}	1.70	85
H_2	4.94×10^{-24}	1.70	78
H_3	4.95×10^{-24}	1.65	74

TABLE I: The values of the minimum percentage error for $H_1 - H_3$ and at which ω and E_{max} they each achieve this.

5. CONCLUSION

Within section 3 the established results of Klco and Savage [16] are reproduced for the JLP, FDC and Harmonic oscillator (HO) basis descriptions of a 0+1 dimensional scalar field theory, including the characteristic dependence of the relative error on the tuning parameters ϕ_{max} and ω , and the improvement associated with satisfying the Nyquist-Shannon sampling conditions. We find in agreement with [16] that the JLP method significantly outperforms the FDC method for a simple harmonic oscillator and for an anharmonic oscillator with $\lambda_0 = 32$ the HO basis outperforms the JLP digitisation method. In addition we see reduced errors with increasing qubit counts as expected for increased Hilbert space dimension.

This benchmark established the HO basis as the most suitable starting point for the Hamiltonian Truncation Effective Theory (HTET) formalism of Cohen *et al.* [25] both because it gave the smallest digitisation errors in the anharmonic case and because it provides the natural basis in which the HTET construction is formulated. HTET was then specialised to the 0+1 dimensional anharmonic oscillator in the HO basis and implemented numerically for $\lambda_0 = 1$. In particular the effective Hamiltonians H_1 , H_2 and H_3 were constructed in the truncated basis, scanned over (ω, E_{max}) and compared against independently obtained high precision reference energies.

We find that HTET can reduce the truncation required to achieve a given accuracy in the HO basis, but only within a restricted region of parameter space where the matching corrections exhibit the expected separation of scales and remain dominated by states near the cut-off. Outside this regime, the finite order HTET expansion loses practical control because the ω dependent HO basis causes abrupt changes in the assignment of states across the truncation boundary, producing instability bands in the error across regions of ω and E_{max} .

HTET is therefore not a uniformly reliable correction within the 0+1 dimensional digitised setting, but instead a controlled and potentially useful improvement only when the expansion remains numerically well controlled within the chosen truncated Hilbert space. In this restricted

regime it provides a route to reducing effective Hilbert space cost and therefore qubit cost, without sacrificing accuracy. This demonstrates that improvements beyond naive basis truncation are possible without enlarging the Hilbert space but also that these improvements are not automatic and depend sensitively on what is closely related to the scale separation underlying the effective theory expansion. HTET therefore is best viewed as, in this 0+1 dimensional setting, a regime dependent correction rather than a universally applicable improvement.

6. FURTHER WORK

A natural extension of the work presented here would include a more systematic study of the boundary between the regime in which HTET is numerically well controlled and the regime in which it appears to lose control. Within the current results a breakdown appears as instability bands across (ω, E_{max}) , associated with the transitions of states near to the truncation boundary abruptly changing treatment leading to a loss of practical scale separation required for a controlled HTET expansion. A more detailed mapping of this boundary and the specific states which induce this effect would allow the regime of validity of HTET to be characterised more precisely for the HO basis and could clarify whether reliable criteria can be established a priori for choosing suitable parameters.

In addition, regions of relatively large higher order corrections are observed at low ω (see figure 6), suggesting a distinct regime in which the structure of HTET corrections differs from that at large ω and in which the effective scale separation required for stable matching may be less robust than the low errors alone would suggest. While this behaviour is not explored in detail here it does provide a starting point for further investigation into the validity of HTET for this application.

Closely related to this is the choice of reference Hamiltonian H_0 and the optimisation of the basis. Within the present work the HO basis was adopted as the natural starting point for HTET, with the auxiliary frequency acting as the tuning parameter. Since the observed breakdown depends on this choice of parameter it would be useful to investigate whether alternative choices of H_0 can enhance the stability of the HTET expansion within the digitisation process.

Further extension to higher order terms of HTET corrections and higher energy levels could demonstrate the utility of HTET and whether the observed instability band structure becomes reduced or enhanced with extensions of that seen here.

Comparisons with alternative methods for improving truncations additionally would prove valuable. Within this work HTET is compared exclusively with the uncorrected HO basis truncation. A broader comparison with other truncation improvement strategies would help determine whether the restricted gains here are specific to HTET or reflect a more general limitation of perturbative corrections within a small truncated Hilbert space.

The results presented within this work assume an ideal quantum computer isolating only digitisation errors. Within realistic NISQ implementation additional error sources such as Trotterisation, gate imperfections and noise are present. A detailed study of how these interact would

clarify whether the reduction in qubit cost achieved by HTET within its regime of validity remains significant under realistic conditions.

While HTET does reduce the effective Hilbert space size required for a given accuracy this does not directly imply an overall computational advantage. Any reduction in qubit cost must be balanced against the increased complexity of constructing and implementing the effective Hamiltonian. This may lead to higher gate counts or circuit depth, therefore a resource analysis is required to determine whether HTET provides a net benefit in practical quantum simulations.

Finally moving beyond the 0+1 dimensional setting and applying the same analysis to the 1+1 dimensional scalar field theory would prove as a suitable extension. This would provide a more physically representative test of the methods studied here and would allow discretisation and truncation to be examined in a setting closer to the eventual target of quantum field theory simulation on quantum devices.

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REFERENCES

- [1] H.-Y. Huang, M. Broughton, J. Cotler, S. Chen, J. Li, M. Mohseni, H. Neven, R. Babbush, R. Kueng, J. Preskill, and J. R. McClean, Quantum advantage in learning from experiments, *Science* **376**, 1182–1186 (2022), doi: 10.1126/science.abn7293.
- [2] B. Fauseweh, Quantum many-body simulations on digital quantum computers: State-of-the-art and future challenges, *Nature Communications* **15** (2024), doi: 10.1038/s41467-024-46402-9.
- [3] E. Farhi, J. Goldstone, and S. Gutmann, A quantum approximate optimization algorithm (2014), doi: 10.48550/arXiv.1411.4028, arXiv:1411.4028 [quant-ph].
- [4] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, and S. Lloyd, Quantum machine learning, *Nature* **549**, 195–202 (2017), doi: 10.1038/nature23474.
- [5] S. S. Gill, O. Cetinkaya, S. Marrone, D. Claudino, D. Haunschild, L. Schlote, H. Wu, C. Ottaviani, X. Liu, S. P. Machupalli, K. Kaur, P. Arora, J. Liu, A. Farouk, H. H. Song, S. Uhlig, and K. Ramamohanarao, Quantum computing: vision and challenges, in *Quantum Computing* (Elsevier, 2025) p. 19–42, doi: 10.1016/b978-0-443-29096-1.00008-8.
- [6] O. Lanes, M. Beji, A. D. Corcoles, C. Dalyac, J. M. Gambetta, L. Henriët, A. Javadi-Abhari, A. Kandala, A. Mezzacapo, C. Porter, S. Sheldon, J. Watrous, C. Zoufal, A. Dauphin, and B. Peropadre, A framework for quantum advantage (2025), doi: 10.48550/arXiv.2506.20658, arXiv:2506.20658 [quant-ph].
- [7] M. Q. Hlatshwayo, M. Babel, D. Islas-Sanchez, and K. Georgopoulos, A Technical Review of Quantum Computing Use Cases for Finance and Economics, *Quantum Rep.* **8**, 26 (2026), doi: 10.3390/quantum8010026.

- [8] Y. e. a. Ji, Quantum Deep Learning: A Comprehensive Review, (2026), doi: 10.48550/arXiv.2603.06644, arXiv:2603.06644 [quant-ph].
- [9] H.-Y. Huang, S. Choi, J. R. McClean, and J. Preskill, The vast world of quantum advantage (2025), doi: 10.48550/arXiv.2508.05720, arXiv:2508.05720 [quant-ph].
- [10] G. Buchs, T. L. Beck, R. S. Bennink, D. Claudino, A. Delgado, N. A. Fadel, P. Groszkowski, K. E. Hamilton, T. S. Humble, N. Kumar, A. Li, P. C. Lotshaw, O. Mukkula, R. Takano, A. Saxena, I.-S. Suh, M. Tsuji, R. Van Beeumen, U. Varetto, Y. Wang, K. Yamazaki, and M. P. Johansson, The role of quantum computing in advancing scientific high-performance computing: A perspective from the adac institute, *Future Generation Computer Systems* **182**, 108487 (2026), doi: 10.1016/j.future.2026.108487.
- [11] J. Preskill, Quantum computing in the nisq era and beyond, *Quantum* **2**, 79 (2018), doi: 10.22331/q-2018-08-06-79.
- [12] J. A. Montanez-Barrera, K. Michielsen, and D. E. B. Neira, Evaluating the performance of quantum processing units at large width and depth (2025), doi: 10.48550/arXiv.2502.06471, arXiv:2502.06471 [quant-ph].
- [13] H. Y. Wong, Review of superconducting qubit devices and their large-scale integration (2026), doi: 10.48550/arXiv.2602.04831, arXiv:2602.04831 [quant-ph].
- [14] F. A. Quinton, P. A. S. Myhr, M. Barani, P. C. del Granado, and H. Zhang, Quantum annealing applications, challenges and limitations for optimisation problems compared to classical solvers (2025), doi: 10.1038/s41598-025-96220-2, arXiv:2409.05542 [quant-ph].
- [15] X.-D. Xie, X. Zhang, B. Koczor, and X. Yuan, Advances in quantum computation in nisq era, *Entropy* **27**, 1074 (2025), doi: 10.3390/e27101074.
- [16] N. Klco and M. J. Savage, Digitization of scalar fields for quantum computing, *Physical Review A* **99**, 10.1103/physreva.99.052335 (2019), doi:10.1103/physreva.99.052335.
- [17] S. P. Jordan, K. S. M. Lee, and J. Preskill, Quantum algorithms for quantum field theories, *Science* **336**, 1130–1133 (2012), doi: 10.1126/science.1217069.
- [18] C. E. Shannon, Communication in the Presence of Noise (1949) doi: 10.1109/JR-PROC.1949.232969.
- [19] M. Hogervorst, S. Rychkov, and B. C. van Rees, A cheap alternative to lattice field theory, *Physical Review D* **91**, 10.1103/physrevd.91.025005 (2015), doi: 10.1103/physrevd.91.025005.
- [20] S. Williams, Simulating quantum field theories on near-term quantum devices, *International Journal of Modern Physics A* **40**, 2530010 (2025), doi: 10.3390/e27101074, <https://doi.org/10.1142/S0217751X25300108>.
- [21] J. Ingoldby, M. Spannowsky, T. Sypchenko, and S. Williams, Enhancing quantum field theory simulations on NISQ devices with Hamiltonian truncation, *Phys. Rev. D* **110**, 096016 (2024), doi: 10.1103/PhysRevD.110.096016, arXiv:2407.19022 [quant-ph].
- [22] A. Macridin, P. Spentzouris, J. Amundson, and R. Harnik, Electron-Phonon Systems on a Universal Quantum Computer, *Phys. Rev. Lett.* **121**, 110504 (2018), doi: 10.1103/PhysRevLett.121.110504, arXiv:1802.07347 [quant-ph].
- [23] A. Macridin, P. Spentzouris, J. Amundson, and R. Harnik, Digital quantum computation of fermion-boson interacting systems, *Phys. Rev. A* **98**, 042312 (2018), doi: 10.1103/PhysRevA.98.042312, arXiv:1805.09928 [quant-ph].

- [24] C. W. Bauer, B. Nachman, and M. Freytsis, Simulating collider physics on quantum computers using effective field theories, *Physical Review Letters* **127**, 10.1103/physrevlett.127.212001 (2021), doi: 10.1103/physrevlett.127.212001.
- [25] T. Cohen, K. Farnsworth, R. Houtz, and M. Luty, Hamiltonian truncation effective theory, *SciPost Physics* **13**, 10.21468/scipostphys.13.2.011 (2022), doi: 10.21468/scipostphys.13.2.011.
- [26] J. Sakurai and J. Napolitano, Modern Quantum Mechanics (2017) pp. 66–67, 337–339, doi: 10.1017/9781108587280.

SUMMARY FOR A GENERAL AUDIENCE

A bit stores information as either a 0 or a 1. A quantum bit, or qubit, can exist in a combination of both states. As more qubits are added, the number of possible states grows exponentially, allowing quantum computers to offer advantages for certain problems.

Quantum field theory (QFT) describes fundamental physics at small scales but is difficult to simulate, however they may benefit from simulation on quantum computers. A key challenge is that fields take continuous values, meaning they require infinite resolution: as you zoom in there is always more to zoom in on. This is incompatible with quantum computers, which have finite resources.

To simulate QFT we have to approximate them using a finite number of values, a process called digitisation. This introduces errors. In this work we study these errors by comparing simulated results with known physical quantities.

We investigate Hamiltonian Truncation Effective Theory (HTET) which attempts to account for the states that are removed in the approximation. We reproduce results showing digitisation errors can be made small and demonstrate that, in certain regimes, HTET can reduce the number of qubits required to achieve high precision.

APPENDIX

Appendix A: Calculation of Observables

To assess the accuracy of the numerical results presented in this work, independently obtained high precision reference values were used for the ground state energy of the 0+1 dimensional scalar field theory with quartic self interaction

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2 + \frac{\lambda_0}{4!}\bar{\phi}^4. \quad (\text{A1})$$

These reference values were obtained using an arbitrary precision shooting method applied to the time independent Schrödinger equation. Writing the ground state wave function in terms of the full Hamiltonian with $\bar{\Pi}$ expressed as $-i\frac{d}{d\bar{\phi}}$ the eigenvalue problem can be given as

$$\left[-\frac{1}{2}\frac{d^2}{d\bar{\phi}^2} + \frac{1}{2}\bar{\phi}^2 + \frac{\lambda_0}{4!}\bar{\phi}^4 \right] \psi(\bar{\phi}) = E\psi(\bar{\phi}) \quad (\text{A2})$$

Since the ground state has even parity, the equation was solved on the half line with $\phi \geq 0$ with boundary conditions of

$$\psi(0) = 1, \quad \psi'(0) = 0 \quad (\text{A3})$$

Then for a trial energy E the differential equation was integrated using the arbitrary precision mpmath ODE solver within python. The eigenvalue was then determined by a bisection in energy, using the node counting theorem to bracket the desired state. For the ground state, the target solution is the even parity solution with no nodes on the half line.

Within this implementation the integration was performed up to a finite cut-off of $\phi_{max} = 5$ with precision set to 50 decimal places. The ground state energy was then extracted by iteratively bisecting the energy interval. This procedure was repeated for each coupling used to define the benchmark values given in table II

λ_0	0+1D Ground State Reference Value
1	0.5277361273141083126903510911470162156139395439930
32	0.8597426904455090193559617737854627804109812323184

TABLE II: Values of the ground state energy for two values of the interaction coefficient λ_0 within the 0+1D case.

The original script used to calculate these values was designed by Professor Ben Pecjak and then transcribed into python for this work.

For harmonic oscillators the reference value of the energy eigenvalues is simply given by

$$E_n = \omega\left(n + \frac{1}{2}\right) \quad (\text{A4})$$

Appendix B: JLP Example Hamiltonian

As an explicit example, the matrix representation of the Hamiltonian is constructed for a simple harmonic oscillator system in dimensionless units ($m = \omega = 1$)

$$H = \frac{1}{2}\Pi^2 + \frac{1}{2}\phi^2. \quad (\text{B1})$$

For a 2 qubit system ($n_\phi = 2^{n_Q} = 4$), at the field truncation limit $\phi_{max} = 3.0$, using JLP digitisation, we obtain a field grid of $[-3, -1, 1, 3]$ from the field spacing

$$\delta_\phi = \frac{2\phi_{max}}{n_\phi - 1} = 2. \quad (\text{B2})$$

$\bar{\Pi}^2$ is produced using a quantum Fourier transform of the diagonal momentum space representation of the operator and the Hamiltonian, which is generally non-diagonal, is represented in the field basis as a matrix:

$$\bar{H} = \begin{pmatrix} 4.9626 & -0.3084 & 0.1542 & -0.3084 \\ -0.3084 & 0.9626 & -0.3084 & 0.1542 \\ 0.1542 & -0.3084 & 0.9626 & -0.3084 \\ -0.3084 & 0.1542 & -0.3084 & 4.9626 \end{pmatrix} \quad (\text{B3})$$

Diagonalisation of the matrix yields the four lowest energy eigenvalues:

Energy Level	0	1	2	3
Calculated Energy Eigenvalue	0.6483	1.2183	4.6601	5.3239

Within this work only the ground state energy is used. The corresponding exact continuum value for anharmonic oscillators is derived in Appendix A and compared with the relevant calculated value to quantify the digitisation error at a given ϕ_{max} and n_Q .